AUNG KAUNG MYAT

GRADUATE GEOSPATIAL ANALYST | GIS, REMOTE SENSING & PYTHON

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Visa / Work Authorisation: Stamp 1G

PROFILE

Geospatial analyst with MSc degrees in GIS & Remote Sensing (UCD, First Class Honours) and Natural Hazards Mitigation & Management. Applies GIS, Python/SQL, remote sensing, deep learning and spatial statistics to flood and coastal hazards, satellite environmental monitoring and urban accessibility, using Irish datasets (CSO, GSI, OPW, INFOMAR, Tailte Éireann).

TECHNICAL SKILLS

GIS & Geospatial Software | ArcGIS Pro, ArcGIS Online (Experience Builder, Dashboards, StoryMaps, Survey123), QGIS

Programming & Spatial Data | Python (GeoPandas, Rasterio, PySAL, OSMnx, ArcPy, PyQGIS), SQL, R

Remote Sensing & Machine Learning | Google Earth Engine, ESA SNAP, Deep Learning (U-Net), OBIA, Random Forest

SELECTED PROJECTS

- **Dublin LiDAR Flood Susceptibility Modelling |** Built an AHP-based model on a 2 m LiDAR DTM (28.87M cells); captured 98.08% of OPW historic flood outlines and 99.32% of the CFRAM 10-year 2D flood extent.
- **Dublin Land-Cover Mapping (Deep Learning vs. OBIA–SVM) |** Compared a ResNet-34 U-Net (~98% validation accuracy) with OBIA-SVM on Sentinel-2 imagery, assessing river continuity and shadow-related errors.
- **Dublin Liveability Index |** Developed a four-pillar Composite Liveability Index for 5,076 Small Areas; classed 284 as Very Low liveability and identified 62 with combined transport and healthcare access gaps.
- **Myanmar Township-Level Conflict Analysis |** Analysed ACLED fatalities across 330 townships (Moran's I = 0.428); identified 22 Gi* hotspots and found the top 10% of townships account for ~85% of recorded fatalities.
- **Satellite-Derived Bathymetry (Tramore Bay) |** Applied a Lyzenga log-ratio model in GEE; summer imagery gave $R^2 = 0.86$, RMSE = 0.79 m against INFOMAR multibeam soundings, while winter turbidity degraded it ($R^2 = 0.12$).

Additional Projects: Tramore Bay Coastal Erosion (DSAS) | Dublin Urban Heat (LST) | Dublin Land-Cover Change Detection (PBIA–RF) | UCD Transit Accessibility (Network Analyst) | Irish Wader Habitat Suitability (ModelBuilder) | Myanmar Multi-hazard Risk Analysis.

PROFESSIONAL EXPERIENCE

Demographics Data Intern | Ministry of Immigration and Population, Mandalay, Myanmar | Jul 2024 - Dec 2024
Standardised demographic datasets to support regional population analysis.

Spatial Planning & Design Consultant (Freelance) | Yangon, Myanmar | Dec 2021 - Feb 2024
Coordinated spatial layouts and contractor workflows for commercial projects, liaising with stakeholders to align deliverables with applicable requirements.

Earth Sciences Teaching Assistant | University of Yangon, Myanmar | Dec 2015 - Apr 2016

ACADEMIC ENGAGEMENT

Research Delegate | NCDR, Taipei City, Taiwan | Sep 2020

Research Delegate | RSDC, Kyoto University, Japan | Aug 2020

EDUCATION

MSc in Geographic Information Systems & Remote Sensing | University College Dublin, Ireland | Sep 2025 - Sep 2026
First Class Honours | GPA 3.76/4.20 | Dissertation: LiDAR-based Multi-criteria Flood Susceptibility Benchmarking

MSc in Natural Hazards Mitigation & Management | National Cheng Kung University, Taiwan | Feb 2020 - Feb 2022
GPA 4.21/4.30 | NCKU Distinguished Scholarship & Disaster Prevention Research Center Scholarship

BSc (Hons) in Geology | University of Yangon, Myanmar | Dec 2013 - Dec 2018
GPA 4.75/5.00

CERTIFICATIONS

Geotechnical Eng. & S.I. (GeoCAD, 2018) **| Advanced RS & GIS** (Science Traveller, 2018) **| Geosoft Target** (Geosoft, 2018)